

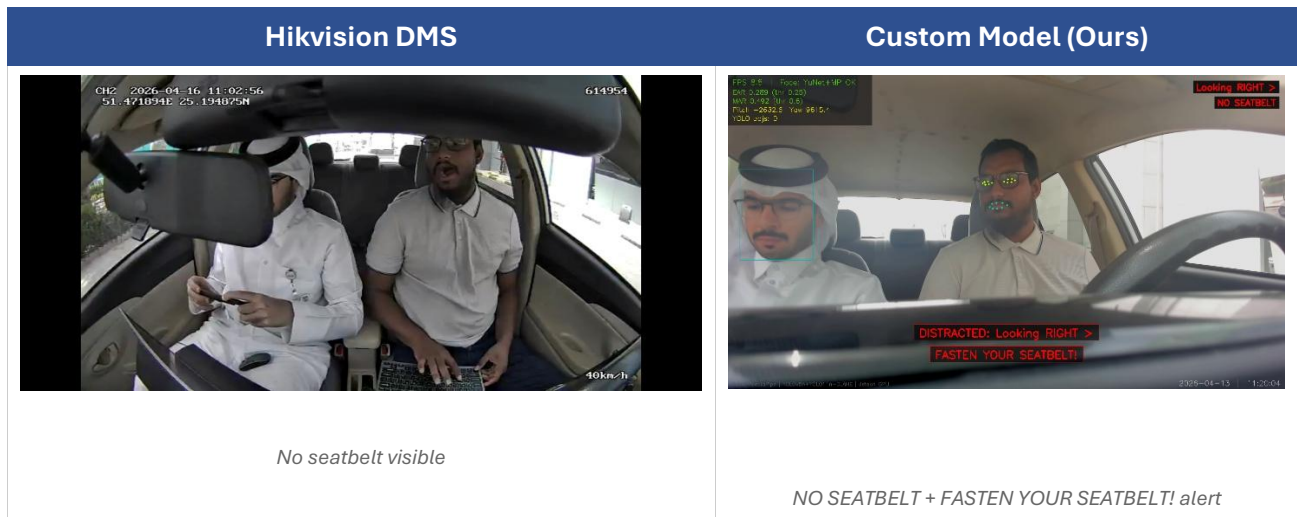
DMS Field Test – Comparison Report

Custom Model vs. Hikvision DMS

This report provides a side-by-side visual comparison of five driver activities detected during a field test. Both systems operated simultaneously in the same vehicle. For each activity captured by the Hikvision camera, the corresponding detection from our custom model is shown.

1. Seatbelt Detection

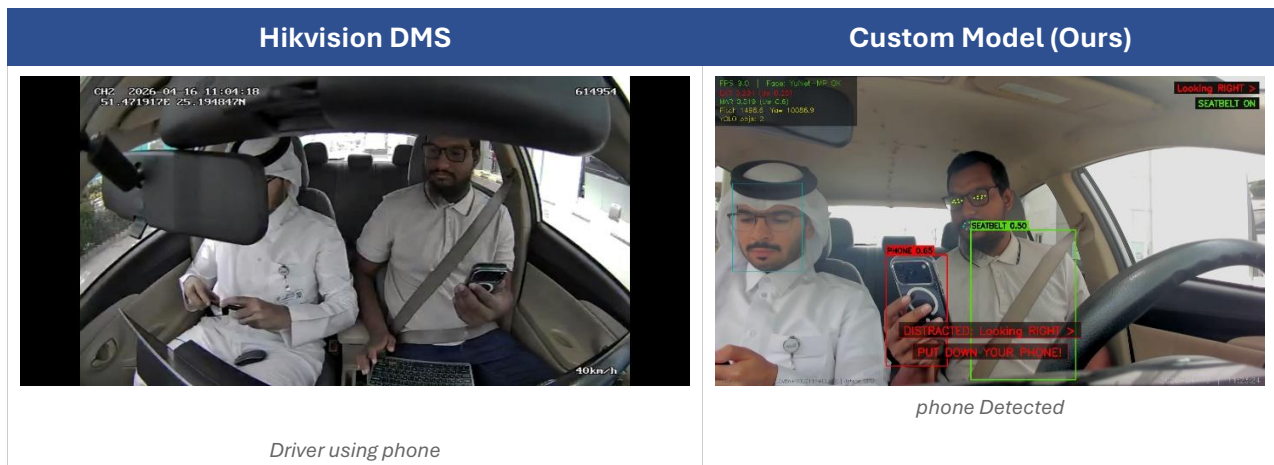
The custom model detects seatbelt presence using YOLO and displays a real-time “NO SEATBELT” / “SEATBELT ON” status. The Hikvision camera captures the scene without on-screen seatbelt alerts.



Result: Both systems captured the seatbelt violation. Our model provides an immediate on-screen alert with status indicator.

2. Phone Use Detection

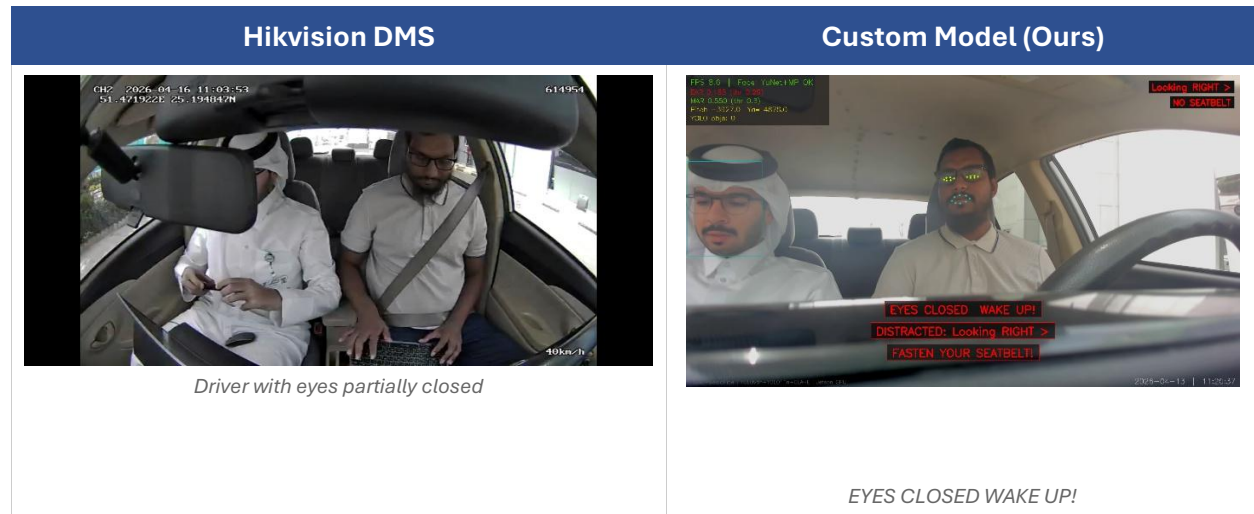
Phone usage is detected by YOLO object detection. The driver was observed holding a phone to his ear in both systems.



Result: Both systems captured phone use. Our model detects it with confidence score (0.53–0.72) but currently labels it as “DRINKING” due to similar hand-to-face posture – label refinement is needed.

3. Eyes Closed (Drowsiness) Detection

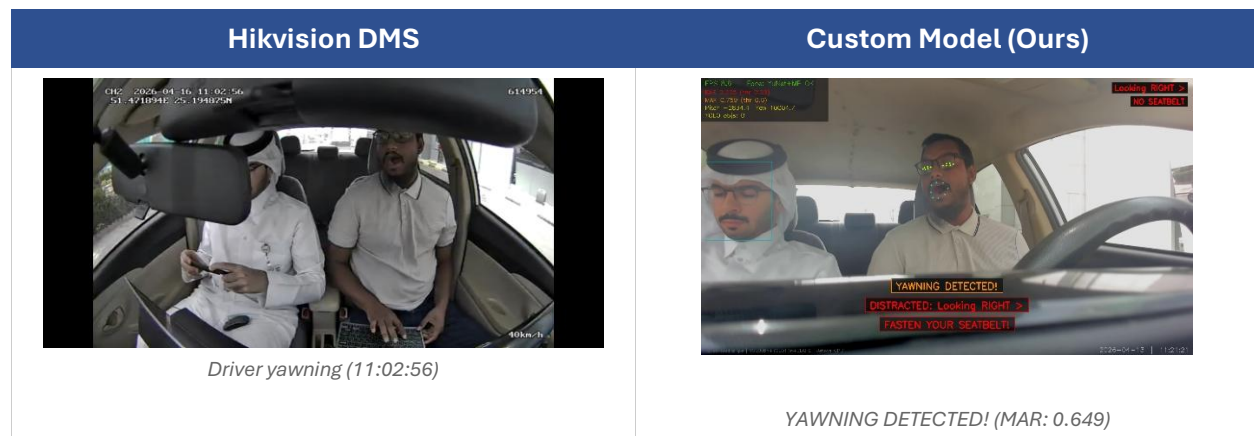
Eye closure is detected using the Eye Aspect Ratio (EAR) from MediaPipe facial landmarks. When EAR drops below 0.25, the alert triggers.



Result: Our model provides precise EAR-based drowsiness detection with immediate alert. Hikvision captures the scene but relies on backend processing for event logging.

4. Yawning (Fatigue) Detection

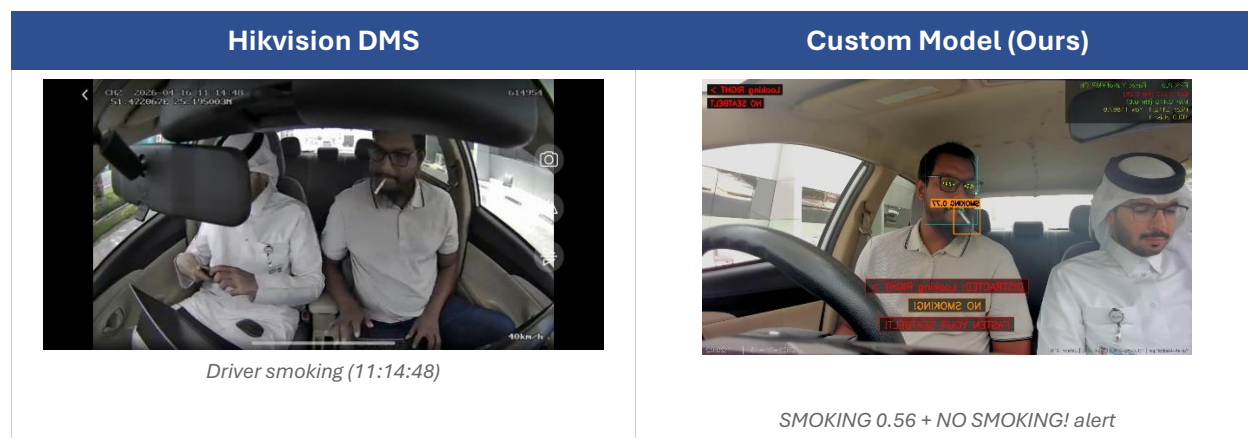
Yawning is detected using the Mouth Aspect Ratio (MAR) from MediaPipe mouth landmarks. Threshold: MAR > 0.6.



Result: Both systems captured the yawning event. Our model triggers a real-time “YAWNING DETECTED!” alert with the MAR value displayed.

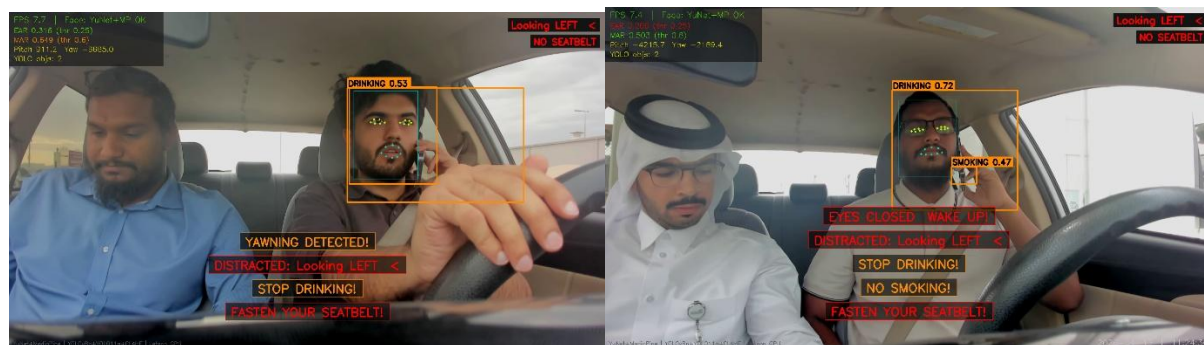
5. Smoking Detection

Smoking is detected by YOLO identifying cigarette-like objects near the driver’s mouth.



Result: Both systems captured smoking. Our model shows a bounding box with confidence score (0.47–0.56) and triggers a “NO SMOKING!” alert in real-time.

6. Errors Custom Model



Summary

Activity	Custom Model	Hikvision
Seatbelt	Detected + Real-time alert	Captured in video
Phone Use	Detected (conf. 0.53–0.72)	Captured in video
Eyes Closed	Detected (EAR < 0.25) + Alert	Captured in video
Yawning	Detected (MAR > 0.6) + Alert	Captured in video
Smoking	Detected (conf. 0.47–0.56) + Alert	Captured in video

Key Takeaway: Our custom model successfully detected all five activities that the Hikvision system captured, with the added advantage of real-time on-screen alerts, confidence scores, and physiological metrics (EAR/MAR). The Hikvision system provides video evidence and In Vehicle Alert.